

Latency-Matched Evaluation Removes the Apparent Cost of Causal Sleep Staging

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Abstract—A sleep-staging model that must run on a wearable device may use only past signal, and the usual assumption is that this constraint costs accuracy. We measure that cost directly, with an ablation in which the same network is trained with and without access to future signal, holding architecture, parameter count, data, folds and seeds fixed. Measured the way this literature normally evaluates, by tiling a recording with non-overlapping context windows, the causal constraint costs 0.025 Cohen’s κ on 78 Sleep-EDF subjects and appears to fall almost entirely on REM. We show that this number is an artifact of the evaluation protocol. Tiling scores predictions at every position of the context buffer, including positions with almost no history, which a streaming system passes through once at start-up and never again; the causal model is crippled there and the non-causal model is not, because it still has lookahead. The reported penalty is arithmetically the mean of a curve over buffer positions, which we verify to within 0.003 κ . Evaluated instead at matched latency, the causal model reaches $\kappa = 0.700$ with zero delay while its non-causal twin cannot exceed $\kappa = 0.698$ at any delay within the context buffer both were trained for, and the apparent REM concentration disappears. We also identify and close a look-ahead leak in the per-epoch normalization used here and in most published baselines; the causal replacement improves both arms, by 0.024 and 0.021 κ , so it was never a trade. The model is strictly causal by construction, verified bit-for-bit from raw sample to logit, emits one label per second, and runs in 30,757 parameters at 0.026 ms per second of EEG on a consumer CPU. Our conclusion is that causality in this setting is a property of the pipeline and of the evaluation protocol, not only of the network, and that measured correctly at both points it is close to free.

Index Terms—sleep staging, electroencephalography, causal models, temporal convolutional networks, self-attention, real-time inference, evaluation methodology

I. INTRODUCTION

Standard clinical polysomnography (PSG) is scored in 30-second epochs by a human technician following American Academy of Sleep Medicine (AASM) conventions [11], and most automated sleep-staging models mirror that unit: they classify one 30-second window at a time and are often free to use signal from later in the recording, or the entire night, to label an earlier epoch. This is a reasonable design for offline analysis of a completed overnight study. It is unsuitable for a wearable device that must report a stage as it happens: for real-time alerting, closed-loop stimulation, or a live dashboard,

a model cannot wait for a later epoch before deciding what occurred a minute earlier.

Causality is a real constraint, so the natural question is what it costs. That question is rarely answered cleanly. Published causal and non-causal staging models differ in capacity, data, preprocessing and optimization all at once, and the gap between them is not attributable to causality in particular. We answer it directly with a matched ablation: the same architecture, parameter count, data, folds and seeds, with only the left-padding and the attention mask removed.

The first result of that experiment is the one we expected. Under the evaluation protocol this literature normally uses, the causal constraint costs 0.025 κ , consistently across seeds and folds. The second result is that this number does not survive contact with a deployment-realistic protocol, and understanding why turns out to be the more useful contribution.

Our contributions are as follows.

First, we identify and close a look-ahead leak in the preprocessing stage of this work and of most published baselines. Per-epoch z -scoring normalizes an early sample using statistics computed over the whole epoch, including its future. Replacing it with a trailing-window statistic, computable one sample at a time, makes the path from raw sample to logit end-to-end causal, verified bit-for-bit. The replacement improves both arms of the ablation, by 0.024 κ causal and 0.021 κ non-causal, so it is a modelling gain rather than the removal of a crutch. Because the leak is close to universal in this literature and costs accuracy rather than providing it, this is a result about the field’s preprocessing conventions and not only about our own pipeline.

Second, we show that the standard batch-tiling evaluation protocol systematically penalizes causal models. Tiling a recording with non-overlapping context windows scores predictions at every position of the buffer, and a causal model at the start of a window has almost no history. A streaming system occupies those positions once, at start-up. We verify that the tiled κ is arithmetically the mean of a curve over buffer positions to within 0.003 for both arms, and that the entire measured penalty is generated by positions with under roughly 60 seconds of history.

Third, we re-measure the cost at matched latency. With a fixed 120-second buffer, the causal model reaches $\kappa = 0.700$ at zero latency, and the non-causal model’s best achievable score is $\kappa = 0.698$, at 37 seconds of delay. Under a full streaming evaluation over every second of every held-out recording, the causal arm scores $\kappa = 0.700$ against the non-causal arm’s 0.693. The penalty is gone.

Fourth, we give the latency–accuracy curve itself, which states what lookahead is worth in seconds of permitted delay. The non-causal arm overtakes the causal arm only beyond about 27 seconds of latency, and even at its optimum does not exceed what the causal model achieves in real time.

Fifth, we show that the apparent concentration of the penalty in REM is the same artifact. Under tiling, REM loses 0.060 F1, four times the next-largest class. Under the deployment protocol the difference is +0.002, and at zero latency the causal model is ahead on REM by 0.025. REM is the class most dependent on context, so it is the class most damaged by an empty buffer.

Sixth, the model itself: a strictly causal staging network whose causality is verified by perturbation test rather than asserted by construction, emitting one label per second in 30,757 parameters, together with a parameter-free causal smoothing step that adds 0.010 κ at inference with no retraining.

II. RELATED WORK

Automated sleep staging from single- or few-channel EEG has moved from hand-engineered spectral features to end-to-end deep learning over the past decade. Supratak et al. proposed DeepSleepNet [1], which pairs a dual-branch CNN with a bidirectional LSTM modelling transition rules across epochs. It reports strong accuracy on Sleep-EDF but reads both past and future epochs through its bidirectional recurrence and uses roughly 22 million parameters. AttnSleep [2] replaces the recurrent stage with multi-head attention over a sequence of epochs, cutting parameter count by about three orders of magnitude while remaining competitive; its attention is not masked, and its 30-second epoch length makes it a batch rather than a streaming method. CareSleepNet [3] is a more recent hybrid convolution–transformer that additionally fuses EEG and EOG, at roughly 22.7 million parameters, and is likewise not causal. Other established single-channel architectures include SeqSleepNet [15], TinySleepNet [14], the single-epoch CNN of Sors et al. [18], SleepEEGNet [16] and IITNet [17]. All of these read future signal, label at 30-second granularity, or both.

A. High-Frequency Staging Is Already Solved, Offline

U-Sleep [13] is the closest prior work on output resolution, and the clearest statement of why resolution by itself is not the contribution here. Trained and evaluated on 15,660 participants across 16 clinical studies, it reports classification performance at fourteen output frequencies spanning 2 to 7,680 stages per minute, the latter being one label per sample at 128 Hz. Its predecessor U-Time [12] introduced the mechanism: an encoder–decoder classifies every input sample, and a small

segment classifier aggregates those classifications over whatever interval is asked for, so the training label rate and the inference output rate are independent. High-frequency sleep staging is therefore not an open problem, and it has not been one since 2019.

It is, however, solved non-causally. U-Sleep is a bidirectional encoder–decoder whose receptive field spans up to 13.5 minutes, roughly half of which lies after the second being labelled, and it makes no streaming or real-time claim. The question this paper asks is what the same resolution costs when the model may not look forward at all.

U-Sleep also supplies the motivation for asking. Generating stages at those fourteen frequencies and testing how well the resulting hypnograms separate obstructive sleep apnea patients from healthy controls, the authors report mean F1 rising from 0.60 at 2 stages per minute to 0.94 at 1,280, and conclude that high-frequency stages are clinically informative. Fine-grained staging is worth having. Whether it can be had causally, and at what price, is what we measure.

B. Causal and Real-Time Staging

A separate line of work targets causal or real-time operation directly. Bresch et al. [7] apply recurrent deep networks to real-time single-channel staging. Pan et al.’s CausalAttenNet [6] is a fast causal long-term-temporal attention network for single-channel EEG, and reports N1 as its weakest class, consistent with the pattern observed here. Mehrotra et al.’s SnoozeNet [8] applies causality-driven transformers to interpretable staging, and Liu et al.’s Micro SleepNet [9] targets real-time staging on mobile and wearable hardware. These establish that network-level causal staging is an active and viable direction, and this work does not claim priority over them on that point.

Streaming causal inference over raw EEG is also active beyond sleep staging. CaMBRAIN [23] applies a causal state-space model with a persistent hidden state to artifact, abnormality, seizure and stress detection at 16 predictions per second, reporting large efficiency gains over windowed baselines. It does not address sleep staging and does not compare against a matched bidirectional control, so it establishes the direction rather than the measurement attempted here.

To our knowledge none of the above reports a matched ablation in which the identical architecture is trained with and without access to future signal on identical folds and seeds, which is what licenses attributing a performance gap to causality rather than to capacity or data. Bresch et al. [7] adopt causal recurrence by design and report no non-causal counterpart; Micro SleepNet [9] ablates its attention and dilation components but not causality, and declines sequence-level modelling outright on memory grounds. We have not been able to inspect the methods sections of [6] and [8] directly, so we state this as a limit of our reading of this literature rather than as an exhaustive claim.

C. Two Points That Cut Across These Groups

The first concerns preprocessing. Causality is a property of a pipeline, not only of a network, and per-epoch normalization,

which computes an epoch’s mean and standard deviation over all 30 seconds and applies them to each sample within it, silently reads the future. It is close to universal in this literature. Section III-D states what it does and what closing it costs; the answer is that closing it is free.

The second concerns evaluation. Tiling a recording with non-overlapping context windows, the default in every implementation we have examined including our own earlier work, scores a model at buffer positions a streaming system does not occupy after start-up. Sections V-C and VI-C show that this alone accounts for the causality penalty we and, by implication, others would otherwise report.

Temporal convolutional networks with causal, dilated convolutions have been shown to match recurrent networks on long sequence-modelling tasks while remaining strictly forward-looking [10]; that building block is adopted here and combined with causally masked attention [19], giving long context without the loss of parallelism that recurrence introduces.

III. DATA AND PREPROCESSING

A. Dataset

This work uses the Sleep-EDF Expanded dataset (sleep-cassette subset) distributed by PhysioNet [5] and described by Kemp et al. [4]: a single EEG channel (Fpz-Cz) resampled to 100 Hz, five AASM-style classes [11] (Wake, N1, N2, N3, REM; N3 and N4 merged per convention). Two subsets are reported: Sleep-EDF-20 (20 subjects, 39 recordings) and Sleep-EDF-78 (78 subjects, 153 recordings, 1,629 hours after Wake trimming), which spans ages 25–101 and is the harder, more representative population. All splits use subject-wise 5-fold cross-validation, verified by unit test (Table IV) so that no subject’s data appears in both a training and a held-out fold.

B. Wake Trimming

Sleep-EDF cassette recordings run approximately 20 hours per recording, and most subjects contribute two consecutive night recordings, so a subject supplies roughly 40 hours of which only a fraction is sleep. Most of the remainder is awake time outside the sleep period, and Wake constitutes 68% of the raw data if retained in full. Each night is trimmed to 30 minutes of Wake on either side of that night’s sleep period, applied per recording before a subject’s two recordings are concatenated, which also removes the daytime gap between them. Trimming reduces the 153 recordings of Sleep-EDF-78 from roughly 3,000 hours to the 1,629 hours reported throughout this paper; every hour figure quoted here is post-trim. This reduces Wake to approximately 15% of the data, matches the convention used by the baselines in Section VI-K, and materially changes which stages the model is rewarded for classifying correctly (Section VI-J).

C. Label Supervision

Expert labels are per-30-second epoch, as is standard for AASM scoring [11]; no public dataset currently provides genuinely per-second expert annotation. Each epoch’s label is

replicated across its 30 constituent one-second training targets, so the network is supervised at 1 Hz even though the ground truth is coarser. Per-second accuracy should therefore be read as an approximation, κ , and accuracy re-aggregated to 30 s by majority vote, are the more comparable figures against 30-second baselines, and both are reported throughout.

D. Normalization: Closing a Look-Ahead Leak

Earlier versions of this pipeline z -scored each 30-second epoch using that epoch’s own mean and standard deviation. The network is causal with respect to its input tensor, but the pipeline that produces that tensor is not: a normalized sample near the start of an epoch depends on raw samples up to 30 s later in the same epoch. The leak is bounded, since it never crosses an epoch boundary, but it is real, and it is inherited from the convention used by the baselines this work compares against.

Closing it requires replacing the per-epoch statistic with a trailing statistic computable one sample at a time, in which the statistics at sample t are drawn only from a preceding window of w seconds and reset at the start of every recording:

$$\mu_t = \frac{1}{w} \sum_{i=t-w+1}^t x_i \quad (1)$$

$$z_t = \frac{x_t - \mu_t}{\sigma_t + \epsilon} \quad (2)$$

where x_t is the raw sample at time t , w is the window in samples, σ_t is the trailing standard deviation over the same window, and ϵ is a small constant guarding against division by zero. The statistic resets at the start of each recording, so the opening seconds run on a partial window: before a full window has been observed ($t < w$) the statistics are taken over the samples seen so far. That is deliberate. A device powering on has no history, and the behaviour is bounded, since within a window of n samples the largest attainable standardized magnitude is $\sqrt{n-1}$, so neither an infinity nor a NaN can arise there. This is mathematically a running statistic a device could compute online, and `src/data/normalization.py` ships both the offline array implementation and the sample-at-a-time filter; they agree to 5.2×10^{-14} over 24,000 samples, so the equivalence is checked rather than asserted.

Every result reported in this paper is produced under (1)–(2). The archived per-epoch runs are retained as a controlled comparison rather than as the headline, because both pipelines were trained from scratch with identical subjects, folds, seeds, architecture and hyperparameters, so the difference between them isolates the normalizer.

Closing the leak improves the model. Cohen’s κ rises from 0.6406 to 0.6649 for the causal arm and from 0.6693 to 0.6898 for the non-causal one under the tiled protocol, and every headline metric moves in the same direction (Table I). A trailing window tracks amplitude drift within an epoch instead of imposing one scale across all 30 seconds, which plausibly explains the gain. The important consequence is that neither

TABLE I
EFFECT OF CLOSING THE PREPROCESSING LEAK (SLEEP-EDF-78, C1, 15
PAIRED RUNS PER ARM, TILED EVALUATION)

	Epoch z -score	Trailing window	Δ
Causal κ	0.6406	0.6649	+0.0243
Non-causal κ	0.6693	0.6898	+0.0205
Causal accuracy	0.7226	0.7505	+0.0279
Causal macro-F1	0.6616	0.6894	+0.0278
Causal N1 F1	0.398	0.421	+0.023
Causal κ at 30 s	0.6515	0.6847	+0.0332
Pipeline reads the future	yes	no	—

arm depended on the leaked information: removing it did not expose a hidden reliance, it removed a source of error. A pipeline step that reads the future is therefore not buying accuracy in exchange for causality here, which is worth stating plainly given how widely the step is used.

IV. MODEL ARCHITECTURE

The network maps a window of raw, single-channel EEG to one five-class prediction per second in four stages: a multi-resolution causal CNN (MRCNN), a causal TCN, causal masked self-attention, and a linear classifier (Table II).

A. Multi-Resolution CNN

Two causal convolution branches read the same raw signal in parallel: a 50-sample kernel (0.5 s at 100 Hz) targeting short events such as spindles and K-complex components, and a 400-sample kernel (4 s) targeting slower activity such as slow waves. Each branch produces 16 channels; concatenation gives 32 features per raw sample, and max-pooling with kernel and stride 100 reduces the 100 Hz representation to one feature vector per second. Pooling is left-aligned, so only present and past samples enter each pooled output.

B. Temporal Convolutional Network

Causality in each convolutional layer is enforced by left-padding only, so that an output depends solely on present and past input:

$$y_t = \sum_{k=0}^{K-1} w_k \cdot x_{t-kd} \quad (3)$$

where K is the kernel size, d the dilation, and the input is left-padded by $(K-1) \cdot d$ zeros so that y_t never reads x_i for $i > t$. Three residual blocks [21] of causal dilated convolutions (dilations 1, 2, 4; kernel 3; two convolutions per block; 32 channels throughout) follow this pattern, each with batch normalization [22], ReLU, dropout (0.2), and a residual connection. Dilation widens the causal receptive field without a larger kernel: the three blocks give a 29-second causal receptive field on the per-second feature stream, on top of the 4-second raw-signal window of the MRCNN's long branch.

Batch normalization pools statistics over the time axis during training, which is not a causal operation. At inference the layers use frozen running statistics and are pure affine

TABLE II
TENSOR FLOW (B = BATCH SIZE, L = CONTEXT LENGTH IN SECONDS)

Stage	Operation	Output shape
Dataset	Load L one-second segments	$(B, L, 100)$
Input	Flatten to one signal	$(B, 1, L \cdot 100)$
MRCNN	Short/long-scale causal convs.	$(B, 32, L)$
TCN	Causal dilated residual blocks	$(B, 32, L)$
Attention	Causal masked self-attention	$(B, L, 32)$
Classifier	Linear, 5 classes	$(B, L, 5)$

TABLE III
PARAMETER BUDGET

Component	Parameters
MRCNN branch, kernel 50 (0.5 s)	848
MRCNN branch, kernel 400 (4 s)	6,448
TCN residual block 1 (dilation 1)	6,336
TCN residual block 2 (dilation 2)	6,336
TCN residual block 3 (dilation 4)	6,336
Causal masked self-attention (4 heads)	4,224
Layer normalization	64
Linear classifier (32 $\rightarrow$ 5)	165
Total	30,757

maps, so the deployed model is causal and the perturbation test of Table IV passes on it. We note this because the training-time behaviour is easy to mistake for a violation of the claim.

C. Causal Self-Attention and Classifier

A four-head attention layer [19], followed by a residual connection and layer normalization, lets each second attend to earlier seconds within the context window:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M\right)V \quad (4)$$

where $M_{ij} = 0$ for $j \leq i$ and $M_{ij} = -\infty$ for $j > i$, so query position i cannot attend to key position j in the future. Attention therefore extends the effective history to the full context window (120 s in the primary configuration) while remaining strictly backward-looking. A final linear layer maps the 32-dimensional representation at each second to five class logits; softmax is not applied inside the model, since the training loss consumes logits directly. Table III gives the parameter budget.

D. Enforcing and Ablating Causality

Causality is enforced in exactly two places: every causal convolution pads only the left side by $(K-1) \cdot d$, per (3), and the attention mask M in (4) zeros out any position after the query. Setting the model to non-causal mode removes both, so that convolutions pad symmetrically and no attention mask is applied, while every layer, channel width, and parameter remains identical. This makes the comparison in Sections V-C and VI-C a controlled one: the two arms differ only in whether the network can access future signal, not in capacity.

Table IV reports the verification suite. Unlike the corresponding table in earlier versions of this work, its scope is

TABLE IV
VERIFIED PROPERTIES (REPOSITORY UNIT TESTS)

Property	Result
Causal conv. preserves sequence length	PASS
Normalization ignores future samples	PASS (bit-identical)
Offline arrays equal the online filter	PASS (5.2×10^{-14})
Raw sample $\rightarrow$ normalization $\rightarrow$ logits	PASS (bit-identical)
Probe does perturb later outputs	PASS (non-vacuous)
Non-causal control does leak	PASS
Incremental buffer equals offline eval.	PASS (79,440 s, 0 diff.)
Output shape and parameter count	PASS (30,757)
Hold-out splits disjoint and complete	PASS
Every subject tested once over 5 folds	PASS
Same seed reproduces the same split	PASS

the whole path and not only the network: the perturbation test runs from raw sample through normalization to logits, and a separate test streams a complete raw recording one sample at a time through the deployment front end and requires the resulting labels to match the offline evaluation exactly. Over one full held-out recording of 79,440 seconds there were zero mismatches.

V. EXPERIMENTAL SETUP

A. Configuration Registry

Three architectural configurations recur throughout the ablations. **C0** is 60 s of context with three TCN blocks. **C1** is 120 s of context with three TCN blocks, selected by the context-length ablation (Section VI-H) and used for every Sleep-EDF-78 result. **C2** is 120 s of context with four TCN blocks, the deeper variant ruled out by the same ablation.

B. Training

Training uses subject-wise 5-fold cross-validation so that no subject’s recordings cross the train/validation/test boundary. The objective is focal loss [20]

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (5)$$

with focusing parameter $\gamma = 2.0$ and class weights α_t , chosen over a weighted sampler for class imbalance, since combining both pushed the model toward over-predicting N1 and reduced its precision to 0.12–0.20 (Section VI-J). Optimization uses a learning rate of 0.001, weight decay 1×10^{-4} , batch size 64, early stopping (patience 8 epochs) on validation κ , and a 40-epoch cap; validation typically plateaus by epoch 10. Checkpoints are selected on validation κ and not macro-F1, since the two peak at different epochs. The held-out test fold is used for neither parameter updates, early stopping, nor checkpoint selection.

For the causality comparison the full 5-fold protocol is repeated under three random seeds (42, 43, 44) on Sleep-EDF-78 for each arm, giving 15 seed-fold result cells per arm.

C. Evaluation Protocols

Two evaluation protocols appear in this paper and the distinction between them is the subject of Section VI-B.

Tiled. The recording is covered by non-overlapping context windows and every position of every window contributes one scored prediction. This is the default in every implementation we have examined, including our own earlier work. Each window begins from a zero-padded empty history.

Streaming. The model recomputes every s seconds against a full buffer and only the freshest s predictions of each window are kept, so after the opening window every scored second is decided with at least $L_{\text{buf}} - s$ seconds of genuine history. This is what a deployed system does. We use $s = 30$ throughout. Every second of every held-out recording is scored exactly once under both protocols, so the two are directly comparable.

D. Statistical Analysis

The causality comparison is repeated under three random seeds. In the archived runs each seed reseeds both the weight initialization and the subject partition, so the experiment is three independent repeats of five-fold cross-validation over fifteen distinct train/test partitions drawn from the 78-subject pool, rather than five partitions measured three times. This is worth stating precisely because it is easy to get wrong: fold 0 of seed 42 and fold 0 of seed 43 share 4 of their 16 test subjects. They are not the same fold.

Because repeats reuse subjects, the fifteen paired differences are not independent, and a naive paired t -test over them overstates significance. The primary test is therefore the Nadeau–Bengio corrected resampled t -test, which inflates the variance estimate by the train-set overlap between repeats, taken as $n_2/n_1 = 1/4$ for five-fold cross-validation. We report the naive statistic alongside it, and a seed-level test at $n = 3$ that treats each repeat as one observation, as the most conservative reading available.

The pairing itself is exact. Within a seed, the causal and non-causal arms use byte-identical subject lists, verified against the archived fold reports, so every one of the fifteen differences is a properly matched pair.

Confidence intervals on single-arm metrics are obtained by bootstrapping whole subjects rather than seconds, with 2,000 draws. Seconds within one recording are heavily autocorrelated, so a second-level interval would be far too narrow to mean anything.

VI. RESULTS

A. Headline Results

Table V reports the primary metrics under both evaluation protocols. Sleep-EDF-78 under the streaming protocol is the primary result: it is the larger and harder population, and the streaming protocol is what a deployed system experiences. The model reaches $\kappa = 0.700$ and 77.7% accuracy per second, rising to $\kappa = 0.707$ and 78.2% when predictions are majority-voted to 30 s epochs. The subject-level bootstrap 95% interval on κ is [0.642, 0.686] under the tiled protocol, computed over the 78 held-out subject records.

κ is lower on the 78-subject set than on the 20-subject set, which is expected: Sleep-EDF-78 spans ages 25–101 and is a harder population, and published baselines show the same

TABLE V
HEADLINE RESULTS, SUBJECT-WISE 5-FOLD CROSS-VALIDATION,
FPZ-CZ, CONFIGURATION C1, SLEEP-EDF-78, 15 FOLDS POOLED

Metric	Tiled	Streaming
Accuracy (per second)	0.7505	0.7772
Cohen’s κ (per second)	0.6649	0.6998
Macro-F1 (per second)	0.6894	0.7230
N1 F1	0.4212	0.4629
Accuracy at 30 s	0.7657	0.7823
κ at 30 s	0.6847	0.7065
Fold-to-fold κ s.d.	0.0257	0.0277
End-to-end causal pipeline	yes	yes
Parameters		30,757
CPU inference		0.026 ms/s

TABLE VI
COST OF CAUSALITY UNDER EACH EVALUATION PROTOCOL
(SLEEP-EDF-78, C1, 15 PAIRED FOLDS)

Protocol	Causal	Non-causal	Δ	Causal loses
Tiled	0.6649	0.6898	−0.0249	14/15
Streaming	0.6998	0.6934	+0.0065	7/15

direction, with AttnSleep reporting 84.4% on Sleep-EDF-20 against 81.6% on Sleep-EDF-78.

B. The Evaluation Protocol Accounts for the Causality Penalty

Table VI is the central result of this paper. The same trained weights, scored under the two protocols of Section V-C, give opposite answers to the question of what causality costs.

Moving from the tiled to the streaming protocol raises the causal arm by 0.0350κ and the non-causal arm by only 0.0036 . The per-seed gains for the causal arm are $+0.0348$, $+0.0345$ and $+0.0355$, so this is not seed noise. Under the streaming protocol the causal arm is ahead by 0.0065 and ahead in 8 of 15 paired folds, which we read as a statistical tie rather than a win.

The mechanism is direct. The streaming protocol keeps the freshest predictions of each window, which carry 90–119 seconds of genuine past. Under tiling those same predictions are scored at every buffer position, averaging roughly 60 seconds of history and reaching zero at the start of each window. The causal model has nothing to work with there. The non-causal model at the same position still has up to 119 seconds of lookahead, so it is barely affected. The two effects nearly cancel.

Section VI-C verifies this quantitatively rather than leaving it as an explanation.

C. Accuracy Against Permitted Latency

If the tiled score is an average over buffer positions, then measuring performance at each position separately should both reproduce the tiled number and show where it comes from. It does.

We define permitted latency L : the label for second t is emitted at $t + L$, so the model sees L seconds of future and $L_{\text{buf}} - L$ seconds of past. $L = 0$ is real-time operation. A

TABLE VII
WHERE THE TILED PENALTY COMES FROM (SLEEP-EDF-78, C1, κ)

Buffer position	Causal	Non-causal	Δ
L 0–29 (91–120 s history)	0.6982	0.6930	−0.0052
L 30–59 (61–90 s history)	0.6923	0.6969	+0.0046
L 60–89 (31–60 s history)	0.6753	0.6911	+0.0158
L 90–119 (1–30 s history)	0.5847	0.6752	+0.0904

single forward pass over a window yields a prediction at every position, and position p carries exactly $L_{\text{buf}} - 1 - p$ seconds of lookahead, so one sweep measures every latency.

Averaging the resulting curve over all 120 positions reproduces the tiled score to within 0.003 for both arms: 0.6626 against a measured 0.6649 for the causal arm, and 0.6890 against 0.6898 for the non-causal arm. The tiled protocol is, arithmetically, this average.

Table VII shows where the penalty lives. At buffer positions with 91–120 seconds of history the causal arm is ahead. The entire aggregate penalty is generated by positions with under roughly 60 seconds of history, and the majority of it by positions with under 30.

Fig. 1 gives the full curve. The causal arm declines monotonically from $L = 0$: it cannot read future signal, so a larger L only costs it history, and at $L = 119$, with one second of history, it collapses to $\kappa = 0.317$. The non-causal arm traces an inverted U, gaining lookahead and losing history, peaking at $\kappa = 0.698$ at $L = 37$ s. The non-causal arm overtakes the causal arm only beyond $L \approx 27$ s.

The comparison that matters is between each arm’s best achievable operating point. The causal model achieves $\kappa = 0.700$ at zero latency. The non-causal model’s optimum is $\kappa = 0.698$, at 37 seconds of delay. With a fixed context buffer, lookahead buys nothing that history does not already provide.

There is a reading of this that we find persuasive but cannot prove from these experiments alone. At zero lookahead the non-causal architecture has no advantage left to use, but it still carries symmetric padding and an unmasked attention, both of which spend capacity on positions that contain nothing. Under that reading the causal constraint stops being a handicap and becomes a useful prior when future signal is unavailable at inference. Confirming it would require varying the constraint independently of the deployment protocol, which we have not done.

D. Context Length and the Fixed-Buffer Objection

Section VI-C measures latency within a buffer of fixed length, so permitting L seconds of lookahead necessarily costs L seconds of history. An offline system would not accept that trade: it would enlarge the buffer and take both. Whether the result survives a larger buffer is therefore a fair question, and we address it as far as these weights allow.

For the causal arm the question is answered exactly. Left-padded convolutions, left-aligned pooling and a causal attention mask together make the output at buffer position p a function of inputs $0 \dots p$ only, so scoring position p of a long

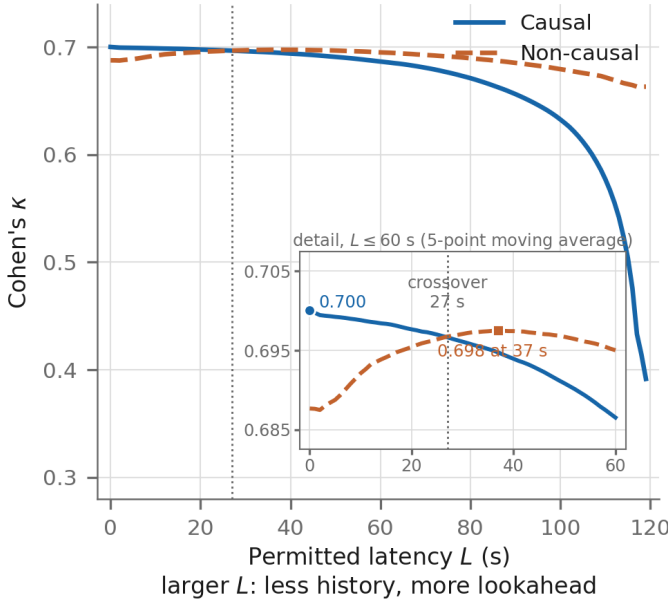


Fig. 1. Cohen’s κ against permitted latency L on Sleep-EDF-78, both arms, 5-point moving average over L . Larger L means less history and more lookahead within a fixed 120 s buffer. The causal arm is best at $L = 0$; the non-causal arm peaks at $L = 37$ s and does not exceed what the causal arm achieves in real time. The causal collapse at large L is the cold-start behaviour that the tiled protocol averages into its score.

TABLE VIII
CAUSAL ARM: κ AGAINST AVAILABLE HISTORY (SLEEP-EDF-78, C1, 360 s BUFFER)

History (s)	30	60	90	120	145	240	360
κ	0.658	0.690	0.701	0.704	0.705	0.698	0.683

window is bit-identical to scoring the last position of a window of length $p+1$. Evaluating one long buffer therefore yields the model’s entire context-response curve at once, and that curve cannot depend on the buffer it was measured in. Table VIII confirms this empirically: measured in buffers of 120, 240 and 360 s, κ at any fixed history agrees to within 0.0035.

The curve rises steeply to about 90 s, is nearly flat from 90 to 180 s, peaks at approximately 145 s, and declines beyond. The optimum is located independently at 145 s in the 240 s buffer and 146 s in the 360 s buffer. Two things follow. A deployed causal system should use a buffer of roughly 145 s rather than the 120 s used here, which is worth about 0.005 κ at no latency cost. And the decline past 180 s is at least partly a train/test length mismatch, since these weights were trained on 120 s sequences; it does not establish that a model trained at longer context would not benefit.

For the non-causal arm the same experiment is not interpretable, and it is worth being explicit about why. Enlarging the buffer necessarily lengthens the sequence its unmasked attention operates over, and unlike the causal arm it has no mask limiting the effective span. Table IX isolates the effect: holding history fixed and increasing only the available lookahead reduces κ monotonically.

TABLE IX
NON-CAUSAL ARM: κ AT FIXED HISTORY AS THE BUFFER GROWS

History	120 s	Buffer 240 s	360 s
60 s	0.6942	0.6714	0.6550
90 s	0.6971	0.6820	0.6666
120 s	0.6870	0.6886	0.6757

TABLE X
PER-CLASS F1 BY EVALUATION PROTOCOL (SLEEP-EDF-78, C1, 15 PAIRED FOLDS). POSITIVE Δ MEANS THE CAUSAL ARM PAYS

Class	Causal	Tiled		Streaming		
		Non-c.	Δ	Causal	Non-c.	Δ
W	0.8931	0.9112	+0.018	0.9039	0.9102	+0.006
N1	0.4212	0.4461	+0.025	0.4629	0.4543	−0.009
N2	0.7728	0.7752	+0.002	0.7964	0.7797	−0.017
N3	0.6654	0.6734	+0.008	0.6970	0.6775	−0.020
REM	0.6945	0.7543	+0.060	0.7547	0.7567	+0.002
Macro	0.6894	0.7120	+0.023	0.7230	0.7157	−0.007

Additional lookahead cannot itself degrade a model that is free to ignore it, so this is the signature of attention operating over sequences longer than those it was trained on rather than a property of lookahead. Consequently the natural test, giving the non-causal arm 120 s of history and 120 s of lookahead in a 240 s buffer, returns $\kappa = 0.689$, which is below the causal arm’s 0.704 at the same history with no lookahead, but which we do not treat as decisive: the figure is depressed by an amount these experiments cannot separate from the comparison.

We therefore state the scope of Section VI-C’s result precisely. Within the buffer both arms were trained for, the causal model at zero latency is not beaten by the non-causal model at any latency. Whether a non-causal model *trained* at a longer context, and so able to use full history and full lookahead together, would exceed it is untested here and is the clearest experiment this analysis calls for. It requires retraining both arms at the longer context and is left to future work.

E. Per-Class Cost

Table X applies the same comparison per class. Under the tiled protocol, REM appears to pay 0.060 F1, four times the next-largest class, and is the only class whose penalty survives Holm correction across the five per-class tests. That result does not survive a change of protocol. Under the streaming protocol the REM difference is +0.002, and at $L = 0$ specifically the causal arm is ahead on REM by 0.025.

The explanation is the same as for the aggregate, and it is mechanistically sensible. REM is defined partly by the context surrounding it, which is why a technician resolves REM against neighbouring epochs rather than scoring it in isolation. It is therefore the class most damaged by a nearly empty buffer, and the cold-start positions that the tiled protocol averages in are exactly where that damage occurs. Fig. 2 shows the two readings side by side.

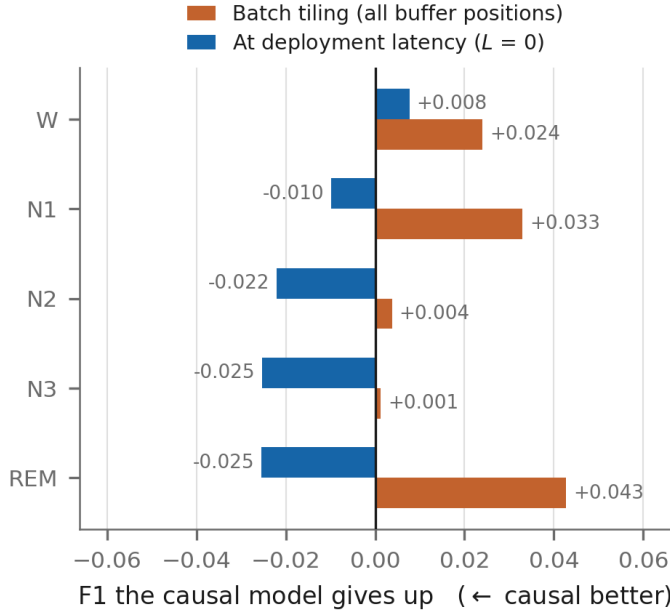


Fig. 2. Per-class F1 the causal arm gives up, under batch tiling (averaged over all buffer positions) and at deployment latency $L = 0$. The apparent REM penalty inverts. Figures are from the latency sweep, which scores each latency on a one-in-five subsample of seconds; Table X gives the full-coverage streaming values.

What survives is a weaker and more careful statement than the earlier one: under tiled evaluation the apparent causality penalty is concentrated in REM, and this is because REM is the class most sensitive to history, not because REM depends on the future in a way the causal constraint forbids.

F. Temporal Stability and Causal Smoothing

Because the model labels every second independently, its raw output is markedly more fragmented than a scored hypnogram. On the Sleep-EDF-78 held-out recordings it changes stage 167 times an hour against 13 times an hour for the technician. A trailing-window mode filter, in which the label at second t becomes the most common prediction over the preceding w seconds,

$$\hat{y}_t = \text{mode}(\{\hat{y}'_{t-w+1}, \dots, \hat{y}'_t\}) \quad (6)$$

uses only past predictions, preserves causality, and requires no retraining (Table XI).

A 30 s window is optimal for κ (+0.0102 over raw) and reduces fragmentation sevenfold; a 45 s window brings the stage-change rate to 18 per hour, close to the human rate, at negligible cost. N1 F1 peaks earlier, around a 10 s window, and declines as smoothing increases, because N1 bouts are short and heavy smoothing absorbs them into neighbouring stages. The optimal window therefore depends on whether overall agreement or transitional-stage sensitivity is prioritized. One caveat applies: the smoothing window was selected on the same held-out folds it is reported on, so the +0.0102 figure is a best-case estimate; selecting w on validation data would be the stricter protocol.

TABLE XI
CAUSAL MODE-FILTER SMOOTHING (SLEEP-EDF-78, C1, STREAMING PIPELINE, SEED 42)

Window (s)	Acc.	κ	M-F1	N1 F1	Chg./hr
1 (raw)	0.7487	0.6634	0.6903	0.414	167
5	0.7514	0.6668	0.6931	0.416	68
10	0.7541	0.6702	0.6960	0.418	53
15	0.7548	0.6711	0.6969	0.417	35
30	0.7568	0.6736	0.6993	0.417	25
45	0.7567	0.6732	0.6993	0.414	18
60	0.7568	0.6732	0.6995	0.411	15
90	0.7549	0.6703	0.6970	0.401	11
120	0.7522	0.6663	0.6937	0.393	9

G. What the Per-Second Output Rate Buys

A per-second output rate is only useful if it lets a system report something sooner. We compare when each system can report a stage change at all, under the causal constraint both face in deployment: the per-second system emits the label for second t at second t , whereas an epoch-level system cannot emit the label for epoch k until epoch k has finished, at second $30(k+1)$. The epoch system therefore carries a structural 30-second floor on any change that begins an epoch.

Both systems are charged for their evidence: a detection requires the new stage to be held for h seconds, so the per-second system reports at the last second of that run rather than the first. Changes the model was already calling during the run-up are excluded, since those are false positives during the outgoing stage rather than early detections, and label changes at a night join are skipped.

At $h = 10$ s, on the conservative subset, the per-second system reports the change first in 94.8% of transitions, with a median 19 seconds saved. The advantage holds across all five stages entered (+19 to +34 seconds).

This result is sensitive to h and it is important to say so. Sweeping h from 3 to 30 seconds, the share of transitions the per-second system reports first falls from 100% at $h = 3$ to 94.8% at $h = 10$, 66.5% at $h = 20$ and 29.6% at $h = 30$, where the epoch system leads. This is mechanically expected: requiring 30 seconds of confirmation reimposes the epoch and discards the resolution. The honest statement is therefore conditional. A finer output rate is worth having precisely when the application is willing to act on less than an epoch of evidence, and is worth nothing when it is not. We report the full sweep rather than a single figure, and note that the scored transition boundary is itself only located to the nearest 30 s, so these numbers describe latency against the scored transition and not against the physiological one.

H. Configuration Ablation

On Sleep-EDF-20, doubling context from 60 s (C0) to 120 s (C1) improved κ by +0.0192; adding a fourth dilated block (C2) changed κ by -0.0050 , within fold variance, so C1 was retained for the Sleep-EDF-78 runs (Table XII). Sequence length changes computation and available history but not

TABLE XII
CONFIGURATION ABLATION, SLEEP-EDF-20, TILED EVALUATION

Config.	Acc.	κ	M-F1	N1 F1
C0 (60 s, 3 blocks)	0.7297	0.6433	0.6713	0.324
C1 (120 s, 3 blocks)	0.7436	0.6625	0.6879	0.336
C2 (120 s, 4 blocks)	0.7405	0.6575	0.6851	0.348

parameter count; the fourth block adds parameters without benefit.

Given the latency result, the context-length finding reads differently than it did before. If history substitutes for lookahead, then more history should continue to help a causal model, and 60 s to 120 s buying 0.019 κ is consistent with that. Whether the curve has flattened by 120 s is untested and is the clearest next experiment.

I. The Small-Sample Warning

Under the tiled protocol on the 78-subject set, the causality effect is -0.0287κ with the causal arm losing in all 15 paired runs, Nadeau-Bengio corrected $t = -4.76$, $p = 0.0003$. The same experiment on the 20-subject split yields $+0.0143\kappa$ with $t(4) = 1.26$, $p = 0.276$: the wrong sign, and not significant. This is not a contradiction. Fold-to-fold variance at 20 subjects is 3.1 times larger than at 78 (κ s.d. 0.092 against 0.030), enough to obscure an effect of this magnitude and even invert it.

We retain this observation even though the headline effect it concerned has now been shown to be a protocol artifact, because the methodological point stands independently and applies to any architectural ablation. Causality penalties measured on ≈ 20 -subject splits, common in this literature for compute reasons, warrant caution, because the true effect can be undetectable or sign-reversed from variance alone.

J. Preprocessing Ablation

Table XIII isolates the effect of wake trimming and the weighted sampler on Sleep-EDF-20. Without trimming, Wake constitutes 68% of the data and inflates accuracy; combining a weighted sampler with focal loss additionally reduced N1 precision to 0.12–0.20. Correcting both improved every individual sleep-stage F1, while overall accuracy decreased because the easy Wake majority previously carrying it was removed. This is a clear case where headline accuracy moves opposite to model quality.

K. Comparison with Published Baselines

Table XIV compares against published single-channel baselines on Sleep-EDF-20, the split on which those baselines report. This model’s rows are its Sleep-EDF-20 results (C1), so the comparison is on a matched subject pool.

DeepSleepNet and AttnSleep remain ahead on both metrics. Both read future signal, and DeepSleepNet uses roughly 700 times more parameters. What this model offers against them is not higher accuracy but a $30\times$ finer output rate at roughly

TABLE XIII
PREPROCESSING ABLATION, SLEEP-EDF-20

Metric	Untrim., sampler on	Trim., sampler off	Δ
Accuracy	0.8064	0.7283	-0.0781
Kappa	0.6584	0.6427	-0.0157
Macro-F1	0.6322	0.6786	$+0.0463$
N1 F1	0.2481	0.3554	$+0.1073$
N2 F1	0.7045	0.7829	$+0.0784$
N3 F1	0.6890	0.7752	$+0.0863$
REM F1	0.5898	0.6784	$+0.0886$
Wake F1	0.9296	0.8010	-0.1286

TABLE XIV
COMPARISON WITH PUBLISHED BASELINES, SLEEP-EDF-20

Model	Acc.	κ	Params	Rate	Causal
DeepSleepNet [1]	82.2%	0.754	$\sim 22\text{M}$	30 s	no
AttnSleep [2] ^a	80.9%	0.709	$\sim 48\text{K}$	30 s	partial
This work, 30 s agg.	76.1%	0.684	30.7K	30 s	yes ^b
This work, per-second	74.4%	0.662	30.7K	1 s	yes ^b

^a In-house rerun at AttnSleep’s native 30-second output over the same 5 folds used throughout this table (80.90%, κ 0.7088). AttnSleep’s own report quotes 84.4% on Sleep-EDF-20 and 81.6% on Sleep-EDF-78 under its own protocol; those figures are not interchangeable with this table’s.

^b Causal through the whole path, verified by the perturbation and streaming-equivalence tests of Table IV.

1/700th the parameter count, with the causal property established rather than assumed, and now with the accompanying result that the causal constraint is not what separates them. Section VI-C bounds the contribution of causality to that gap at approximately zero under matched deployment conditions, which leaves the remainder to capacity, architecture and training scale.

CareSleepNet [3] is discussed rather than tabulated. Its frequently quoted figure of 78.07%/0.694 is its best fold, not its mean; its own report shows fold 1 at 38.12% and fold 4 as unrecoverable. Without access to its complete per-fold results a fair mean-based comparison is not possible, so no head-to-head claim is made here.

L. Efficiency

Latency was benchmarked on an Intel Core i9-14900HX CPU using a corrected protocol: batch size 1, single thread, 100 warm-up and 1,000 timed runs, median and interquartile range reported, with the normalizer measured separately. C1 costs 0.0271 ms per second of EEG and C0 0.0216 ms.

This metric is per-output throughput, not per-invocation latency: it is total wall-clock time for a synthetic sequence divided by its length in seconds of EEG. A device labelling every second pays a rolling cost instead, which is roughly forty times larger and is what `scripts/streaming_demo.py --mode rolling` measures. Absolute per-invocation latency, peak memory, and MAC counts are not measured here and are needed before any claim about a specific target device.

VII. DISCUSSION

The first thing to say is what changed and why. An earlier version of this work reported that the causal constraint costs 0.029κ and that the cost concentrates in REM. Both statements are reproducible under the evaluation protocol they were measured with, and both are artifacts of that protocol. We report them here alongside the corrected measurement rather than quietly replacing them, because the discrepancy is the finding.

The mechanism is not subtle once seen. A streaming model is defined by what it can do with the history it has accumulated. Tiling a recording with independent context windows repeatedly destroys that history and then scores the model on the wreckage, and it does so asymmetrically: a model that can read forward is largely unaffected at the start of a window, and a model that cannot is crippled there. The tiled score is the mean over buffer positions, which we verify to within 0.003κ , and the causality penalty is generated almost entirely by positions a deployed system occupies once and then leaves.

We do not claim from this that causality is free in general. Two things bound the claim. First, the streaming protocol denies the non-causal arm the lookahead that defines it, so this is a comparison of deployment options and not a clean architectural ablation; the fair framing is matched latency, and at matched latency the answer is that lookahead is worth nothing the causal model cannot obtain from history. Given unbounded latency, which is the offline setting, lookahead is worth roughly 0.025κ . Second, the non-causal arm here is our own matched twin rather than a state-of-the-art offline model, so the comparison bounds what causality costs within this capacity class and not in general. Third, the buffer is fixed at the length both arms were trained for. Section VI-D shows that the causal arm’s context-response curve can be recovered exactly from a longer buffer, because masking makes it buffer-invariant, but that the same manoeuvre does not work for the non-causal arm: enlarging its buffer lengthens the sequence its unmasked attention spans, and the resulting degradation cannot be separated from the effect under study. The claim is therefore scoped to the trained buffer, and a non-causal arm trained at longer context is the experiment that would extend or overturn it.

The practical implication is narrow and useful. A system that must operate in real time should not expect to pay for it in accuracy, and should not be evaluated with a protocol that makes it look as though it does. Reported causality penalties in this literature should be read with the evaluation protocol in mind, and we would encourage any comparable ablation to report both protocols, since the two numbers differ by more than the effect being measured.

The preprocessing finding deserves emphasis for the same reason. A network can be causal in every layer, verified bit-for-bit, and still sit downstream of a normalization step that reads the future. That configuration is easy to build accidentally and hard to notice, because the network passes every causality test one would think to write. We found it by verifying our own

work rather than because a reviewer asked, and the verification in Table IV is built to cover the whole path for exactly that reason.

Finally, the per-second output rate. Section VI-G gives the first quantitative answer we are aware of to what a finer rate buys: at a 10-second confirmation window, transitions are reported a median 19 seconds sooner than an epoch-level system can report them, in 94.8% of cases. That advantage is conditional on the confirmation window and disappears by 30 seconds, which is the honest form of the claim and also the useful one: it tells a system designer exactly when the finer rate is worth the fragmentation it brings.

VIII. LIMITATIONS

This work is limited to a single channel (Fpz-Cz) and a single dataset family (Sleep-EDF); generalization to other electrode montages, sensors, or clinical populations is untested, as is any second architecture. The protocol finding in particular is demonstrated on one architecture, and although the mechanism is generic, its magnitude on other models is unknown.

Supervision is 30-second expert labels replicated to 1 Hz, because no public dataset currently provides per-second ground truth, so per-second accuracy should be read as an approximation and κ treated as the more meaningful summary. Section VI-G measures when a per-second system can report a change rather than how accurately it localizes one, precisely because the scored boundary is itself only known to 30 seconds.

No human inter-scorer agreement ceiling is available for Sleep-EDF, so the κ values here are positioned against published models rather than against what a second expert would achieve. A multi-scored dataset would supply that reference.

Two free parameters are untested. The normalization window was fixed at 30 s to match the epoch length it replaces, and no other setting was trained. The context length was fixed at 120 s; Section VI-D shows post hoc that approximately 145 s would have been better for the causal arm, and that the question of whether a non-causal model trained at longer context could use history and lookahead together to exceed the causal arm at zero latency is open. Both require retraining and neither is resolved here.

The causality estimate rests on fifteen paired runs over one dataset family, and those runs reuse subjects across repeats, which is what the Nadeau–Bengio correction accounts for. A design with the partition held fixed across seeds would separate initialization variance from partition variance and we did not have one.

The smoothing window was selected on the same held-out folds used to report it, so its gain is a best-case estimate. Efficiency is characterized by CPU throughput alone; MAC counts, peak inference memory, per-invocation latency, and measurements on an actual wearable-class device are not reported, so the real-time claim should be read as evidence of feasibility on a desktop CPU, not as a device-level result.

IX. CONCLUSION

A strictly causal, per-second sleep-staging model was built, verified end-to-end from raw sample to logit, and compared against a non-causal twin identical in every other respect. Measured with the batch-tiling protocol standard in this literature, the causal constraint costs 0.025κ and appears to fall almost entirely on REM. Measured at matched latency, it costs nothing: the causal model reaches $\kappa = 0.700$ with zero delay, the non-causal model cannot exceed $\kappa = 0.698$ at any delay within the buffer both were trained for, and the REM concentration disappears. The tiled score is arithmetically the mean over context-buffer positions, and the penalty it reports is generated by positions a streaming system occupies once at start-up.

Two further results support the same theme, that causality in this setting is a property of the pipeline and the protocol rather than only of the network. A look-ahead leak in per-epoch normalization, near-universal in this literature, was identified and closed, and closing it improved both arms. And a parameter-free causal smoothing step recovers output stability at inference with no retraining.

At 30,757 parameters and 0.026 ms per second of EEG the model is compact and fast enough for on-device operation while emitting labels $30\times$ more often than epoch-level systems. Whether longer past context continues to substitute for lookahead, and whether the protocol effect reproduces on a second architecture and a second dataset, are the two clearest directions this measurement opens.

REPRODUCIBILITY AND DATA AVAILABILITY

Subject-wise 5-fold splits, per-fold checkpoints, archived per-fold result files, the statistical analysis, the latency sweep, the smoothing sweep, and the boundary-latency analysis are available at <https://github.com/swapnil5053/causal-sleep-staging>, alongside the implementation, configuration files, evaluation scripts, and unit tests covering the causality and split properties reported in Table IV. Per-second predictions and logits are archived for every held-out fold, so every analysis in Sections VI-B–VI-G can be reproduced without retraining. A camera-ready submission will cite an archived snapshot with a persistent identifier and a pinned commit hash rather than a branch reference.

The Sleep-EDF Expanded dataset is publicly available from PhysioNet [4], [5] under the Open Data Commons Attribution License and is not redistributed here. This work is a secondary analysis of that de-identified public dataset and involved no new human-subject data collection, so institutional ethics review was not required.

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DECLARATION OF COMPETING INTERESTS

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